

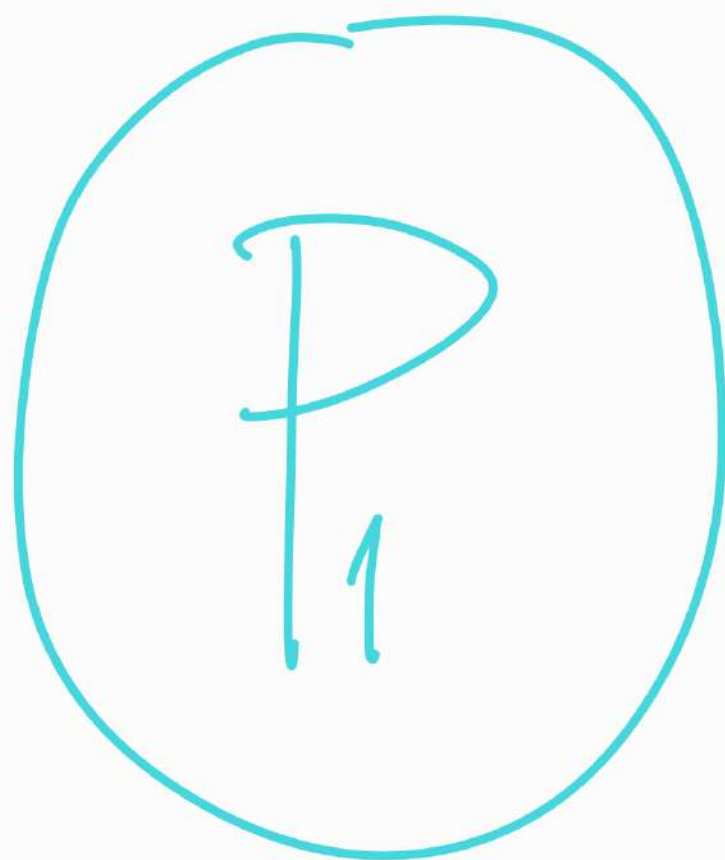
PAUTA AUX 5

POTENCIAL, CONDUCTORES Y CONDENSADORES

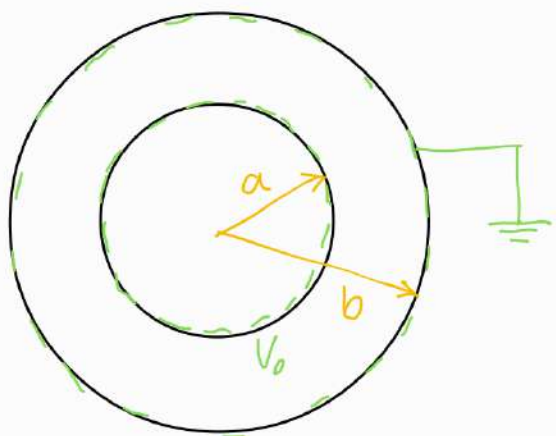
FI2002-2

2024-1

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Auxiliares: Camila M., Bianca Z.
Ayudantes: Julio D., Gerd H.



Esquema del problema:



Se conoce el campo eléctrico, que es definido por partes, para esta distribución de simetría cilíndrica.

Se tiene como dato el valor del potencial en dos radios:

$$V(r=a) = V_0 \quad \text{y} \quad V(r=b) = 0.$$

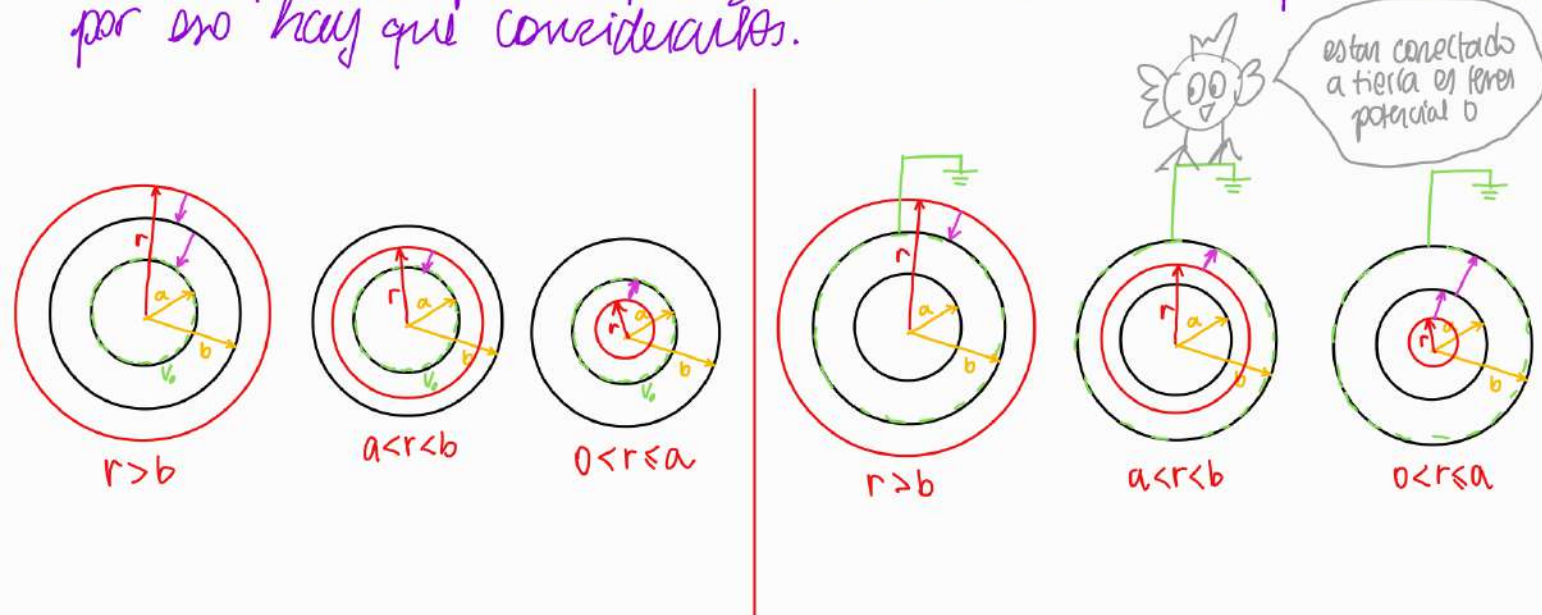
Para calcular el potencial en todo el espacio, siempre se requiere una referencia, tal que nos sea útil.

Como se conoce el valor del potencial ahí, se tomará esa ref.

Ya que el dato es el campo eléctrico, para llegar al potencial se utiliza $V(A) - V(B) = - \int_B^A \vec{E} \cdot d\vec{r}$.

Debido a que el campo eléctrico está definido por partes, el potencial también (en los mismos espacios), pero ¡DEBE ser continuo!

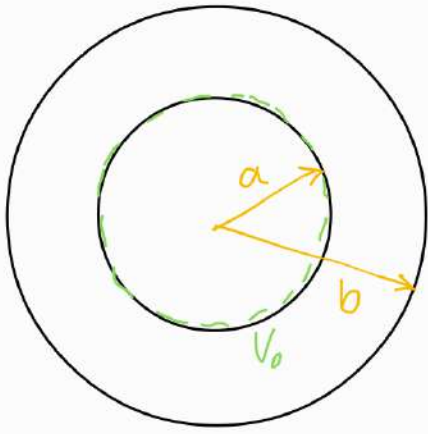
La idea siempre es "acercarse" a la referencia. Al hacer eso, se podría pasar por zonas con distinto campo eléctrico; por eso hay que considerarlo.



(P₁) a) PROPUESTO //

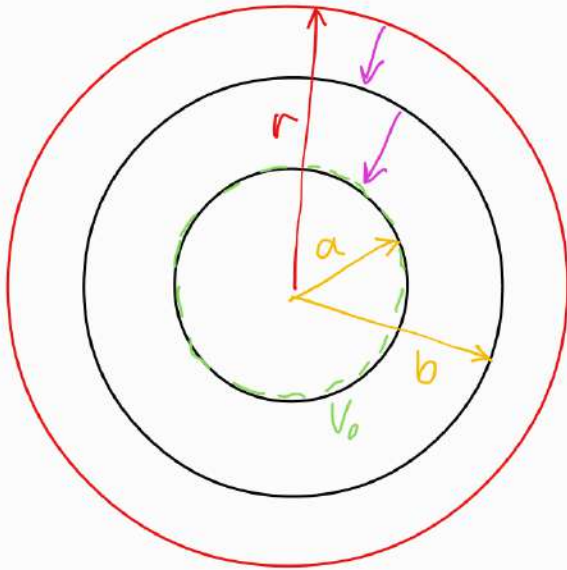
Hacerlo como ejercicio de Ley de Gauss ☺

(P₁) b) Se tiene que : • $V(r_{ref}=a)=V_0$



$$\vec{E} = \begin{cases} \frac{\rho}{2\epsilon_0} r \hat{r} , & 0 < r \leq a \\ \frac{\rho a^2}{2\epsilon_0} \frac{1}{r} \hat{r} , & a < r < b \\ 0 \hat{r} , & r > b \end{cases}$$

$$r > b$$



$$V(r > b) - V(a) = - \int_a^r \vec{E} \cdot d\vec{l}$$

$$= - \left(\underbrace{\int_b^r \vec{E}(r > b) \cdot d\vec{l}}_{\text{I}} + \underbrace{\int_a^b \vec{E}(a < r < b) \cdot d\vec{l}}_{\text{II}} \right)$$

(ver desarrollo más abajo)

$$= - \left(\underbrace{0}_{\text{I}} + \underbrace{\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right)}_{\text{II}} \right)$$

$$\Rightarrow V(r > b) = - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) + V_0 //$$

$$\textcircled{\text{I}} : \int_b^r \vec{E}(r > b) \cdot d\vec{l} \quad \left\| \begin{array}{l} \cdot \vec{E}(r > b) = 0 \hat{r} \\ \cdot d\vec{l} = dr \hat{r} \end{array} \right.$$

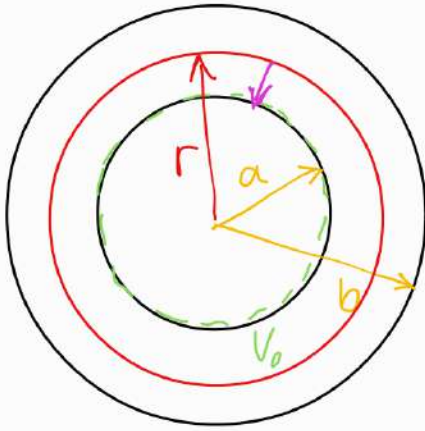
$$= 0 \int_b^r dr = 0 //$$

$$\textcircled{\text{II}} : \int_a^b \vec{E}(a < r < b) \cdot d\vec{l} \quad \left\| \begin{array}{l} \cdot \vec{E}(a < r < b) = \frac{\rho a^2}{2\epsilon_0} \frac{1}{r} \hat{r} \\ \cdot d\vec{l} = dr \hat{r} \end{array} \right.$$

$$= \frac{\rho a^2}{2\epsilon_0} \int_a^b \frac{1}{r} dr = \frac{\rho a^2}{2\epsilon_0} \left[\ln(|r|) \right]_a^b = \frac{\rho a^2}{2\epsilon_0} [\ln(b) - \ln(a)] = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) //$$

$r > 0$
 $\Rightarrow |r| = r$

$$\underline{a < r < b}$$



$$\bullet V(a < r < b) - V(a) = - \int_a^r \vec{E} \cdot d\vec{l}$$

$$= - \underbrace{\left(\int_a^r \vec{E}(a < r < b) \cdot d\vec{l} \right)}_{\textcircled{I} \atop a < r < b}$$

(ver desarrollo
más abajo)

$$= - \underbrace{\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right)}_{\textcircled{I} \atop a < r < b}$$

$$\Rightarrow V(a < r < b) = - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right) + V_0$$

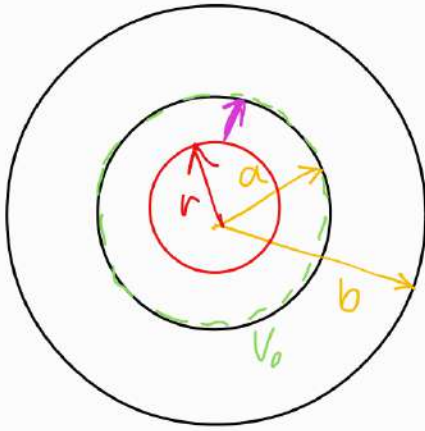
$$\textcircled{I} : \int_a^r \vec{E}(a < r < b) \cdot d\vec{l} \quad \left\| \begin{array}{l} \cdot \vec{E}(a < r < b) = \frac{\rho a^2}{2\epsilon_0} \frac{1}{r} \hat{r} \\ \cdot d\vec{l} = dr \hat{r} \end{array} \right.$$

$a < r < b$

$$= \frac{\rho a^2}{2\epsilon_0} \int_a^r \frac{1}{r} dr = \frac{\rho a^2}{2\epsilon_0} \left[\ln(|r|) \right]_a^r = \frac{\rho a^2}{2\epsilon_0} [\ln(r) - \ln(a)] = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right)$$

$r > 0$
 $\Rightarrow |r| = r$

$$0 < r \leq a$$



$$V(0 < r \leq a) - V(a) = - \int_a^r \vec{E} \cdot d\vec{l}$$

$$= - \left(\int_a^r \vec{E}(0 < r \leq a) \cdot d\vec{l} \right)$$

(I)
 $0 < r \leq a$

(ver desarrollo
más abajo)

$$= - \frac{\rho}{4\epsilon_0} (r^2 - a^2)$$

(I)
 $0 < r \leq a$

$$\Rightarrow V(0 < r \leq a) = - \frac{\rho}{4\epsilon_0} (r^2 - a^2) + V_0$$

$$\textcircled{\text{I}} : \int_a^r \vec{E}(0 < r \leq a) \cdot d\vec{\ell} \quad \left\| \begin{array}{l} \vec{E}(0 < r \leq a) = \frac{\rho}{2\epsilon_0} r \hat{r} \\ d\vec{\ell} = dr \hat{r} \end{array} \right.$$

$$= \frac{\rho}{2\epsilon_0} \int_a^r r dr = \frac{\rho}{2\epsilon_0} \cdot \frac{1}{2} [r^2]_a^r = \frac{\rho}{4\epsilon_0} [r^2 - a^2]$$

Entonces, queda el potencial definido en todo el espacio, con respecto a la referencia en $r=a$:

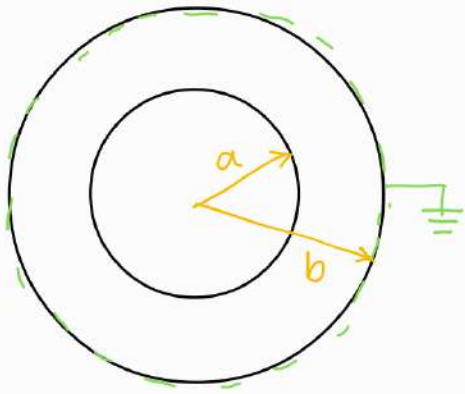
$$V_a(r) := \begin{cases} \bullet V(0 < r \leq a) = -\frac{\rho}{4\epsilon_0} (r^2 - a^2) + V_0 \\ \bullet V(a < r < b) = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right) + V_0 \\ \bullet V(r > b) = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) + V_0 \end{cases}$$

y además, está bien definido pues es continuo:

$$\begin{aligned} \bullet V(0 < r \leq a) \Big|_{r=a} &= -\frac{\rho}{4\epsilon_0} (a^2 - a^2) + V_0 \\ &= -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{a}\right) + V_0 = V(a < r < b) \Big|_{r=a} // \end{aligned}$$

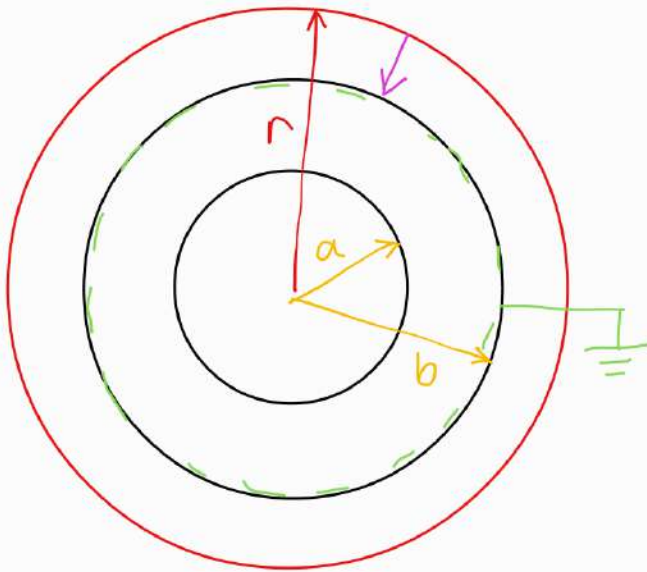
$$\bullet V(a < r < b) \Big|_{r=b} = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) + V_0 = V(r > b) \Big|_{r=b} //$$

(P₁) c) Se tiene que: $V(r_{\text{ref}} = b) = 0$



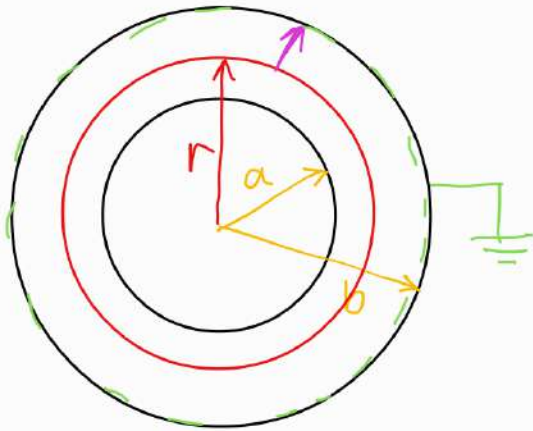
$$\vec{E} = \begin{cases} \frac{\rho}{2\epsilon_0} r \hat{r} , & 0 < r \leq a \\ \frac{\rho a^2}{2\epsilon_0} \frac{1}{r} \hat{r} , & a < r < b \\ 0 \hat{r} , & r > b \end{cases}$$

$$r > b$$



$$\begin{aligned}
 \bullet V(r > b) - V(\cancel{b})^0 &= - \int_b^r \vec{E} \cdot d\vec{\ell} \\
 &= - \int_b^r \vec{E}(r > b) \cdot d\vec{\ell} \quad \left\| \begin{array}{l} \bullet \vec{E}(r > b) = 0 \hat{r} \\ \bullet d\vec{\ell} = dr \hat{r} \end{array} \right. \\
 &= - 0 \int_b^r dr = 0 \\
 \Rightarrow \boxed{V(r > b) = 0} //
 \end{aligned}$$

$$\underline{a < r < b}$$



$$\bullet V(a < r < b) - V(\cancel{b}) = - \int_b^r \vec{E} \cdot d\vec{\ell}$$

$$= - \left(\int_b^r \vec{E}(a < r < b) \cdot d\vec{\ell} \right)$$

$$\textcircled{I}$$

 $a < r < b$

$$= - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{b}\right)$$

$$\textcircled{I}$$

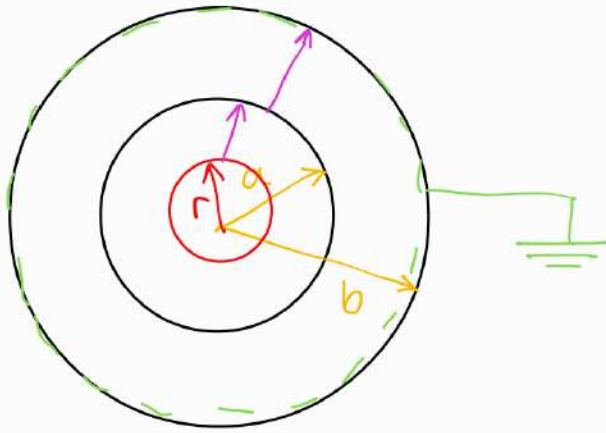
 $a < r < b$

$$\Rightarrow V(a < r < b) = - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{b}\right)$$

$\textcircled{I}$: $\int_b^r \vec{E}(a < r < b) \cdot d\vec{\ell}$
 $\left\| \begin{array}{l} \cdot \vec{E}(a < r < b) = \frac{\rho a^2}{2\epsilon_0} \frac{1}{r} \hat{r} \\ \cdot d\vec{\ell} = dr \hat{r} \end{array} \right.$

$$= \frac{\rho a^2}{2\epsilon_0} \int_b^r \frac{1}{r} dr = \frac{\rho a^2}{2\epsilon_0} \left[\ln(|r|) \right]_b^r \stackrel{r > 0}{\Rightarrow |r| = r} = \frac{\rho a^2}{2\epsilon_0} [\ln(r) - \ln(b)] = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{b}\right)$$

$$0 < r \leq a$$



$$\begin{aligned} V(0 < r \leq a) - V(b) &= - \int_b^r \vec{E} \cdot d\vec{l} \\ &= - \left(\underbrace{\int_a^r \vec{E}(0 < r \leq a) \cdot d\vec{l}}_{\text{I}} + \underbrace{\int_b^a \vec{E}(a < r \leq b) \cdot d\vec{l}}_{\text{II}} \right) \end{aligned}$$

$$= - \left(\underbrace{\frac{\rho}{4\epsilon_0} (r^2 - a^2)}_{\text{I}} + \underbrace{\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right)}_{\text{II}} \right)$$

$$\Rightarrow V(0 < r \leq a) = - \left(\frac{\rho}{4\epsilon_0} (r^2 - a^2) + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right) \right)$$

$$\textcircled{\text{I}} : \int_a^r \vec{E}(0 < r \leq a) \cdot d\vec{l} \quad \left\| \begin{array}{l} \vec{E}(0 < r \leq a) = \frac{\rho}{2\epsilon_0} r \hat{r} \\ d\vec{l} = dr \hat{r} \end{array} \right.$$

$$= \frac{\rho}{2\epsilon_0} \int_a^r r dr = \frac{\rho}{2\epsilon_0} \cdot \frac{1}{2} [r^2]_a^r = \frac{\rho}{4\epsilon_0} [r^2 - a^2] //$$

$$\textcircled{\text{II}} : \int_b^a \vec{E}(a < r < b) \cdot d\vec{l} \quad \left\| \begin{array}{l} \vec{E}(a < r < b) = \frac{\rho a^2}{2\epsilon_0} \frac{1}{r} \hat{r} \\ d\vec{l} = dr \hat{r} \end{array} \right.$$

$$= \frac{\rho a^2}{2\epsilon_0} \int_b^a \frac{1}{r} dr = \frac{\rho a^2}{2\epsilon_0} \left[\ln(|r|) \right]_b^a \quad \left| \begin{array}{l} r > 0 \\ \Rightarrow |r| = r \end{array} \right. = \frac{\rho a^2}{2\epsilon_0} [\ln(a) - \ln(b)] = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right) //$$

Entonces, queda el potencial definido en todo el espacio, con respecto a la referencia en $r=b$:

$$V_b(r) := \begin{cases} \bullet V(0 < r \leq a) = - \left(\frac{\rho}{4\epsilon_0} (r^2 - a^2) + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right) \right) \\ \bullet V(a < r < b) = - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{b}\right) \\ \bullet V(r > b) = 0 \end{cases}$$

Y además, está bien definido pues es continuo:

$$\begin{aligned} \bullet V(0 < r \leq a) \Big|_{r=a} &= - \left(\frac{\rho}{4\epsilon_0} (a^2 - a^2) + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right) \right) \\ &= - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right) = V(a < r < b) \Big|_{r=a} \end{aligned}$$

$$\begin{aligned} \bullet V(a < r < b) \Big|_{r=b} &= - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{b}\right) \\ &= 0 = V(r > b) \Big|_{r=b} \end{aligned}$$

(P1) d) Se tienen los siguientes potenciales

$$V_a(r) := \begin{cases} \bullet V(0 < r \leq a) = -\frac{\rho}{4\epsilon_0} (r^2 - a^2) + V_0 \\ \bullet V(a < r < b) = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right) + V_0 \\ \bullet V(r > b) = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) + V_0 \end{cases}$$

$$V_b(r) := \begin{cases} \bullet V(0 < r \leq a) = -\left(\frac{\rho}{4\epsilon_0} (r^2 - a^2) + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right)\right) \\ \bullet V(a < r < b) = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{b}\right) \quad V_0 = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{a}{b}\right) \\ \bullet V(r > b) = 0 \quad -\frac{\rho a^2}{2\epsilon_0} \left(\ln\left(\frac{r}{b}\right) + \ln\left(\frac{a}{b}\right)\right) \end{cases}$$

Ambos difieren por una constante, pero describen el mismo campo eléctrico en el espacio, ya que si $V_a = V_b + \beta$, $\beta \in \mathbb{R}$ constante

$$\Rightarrow \nabla V_a = \nabla(V_b + \beta) = \nabla V_b + \nabla \beta \overset{\rightarrow 0}{\underset{\text{es constante}}{\downarrow}} \Rightarrow \vec{E} = -\nabla V_a = -\nabla V_b //$$

Notar que dicha constante es $\beta = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) - V_0$.

En efecto:

$$\begin{aligned} \bullet V_a(r > b) + \beta &= -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) + V_0 + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) - V_0 \\ &= 0 = V_b(r > b) // \end{aligned}$$

$$\begin{aligned} \bullet V_a(a < r < b) + \beta &= -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right) + V_0 + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) - V_0 \\ &= \frac{\rho a^2}{2\epsilon_0} \left(\ln\left(\frac{b}{a}\right) - \ln\left(\frac{r}{a}\right) \right) = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{r}\right) \\ &= \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{r}\right) = -\frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{r}{b}\right) = V_b(a < r < b) // \end{aligned}$$

$$\begin{aligned} \bullet V_a(0 < r \leq a) + \beta &= -\frac{\rho}{4\epsilon_0} (r^2 - a^2) + V_0 + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) - V_0 \\ &= -\frac{\rho}{4\epsilon_0} (r^2 - a^2) + \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) \\ &= -\frac{\rho}{4\epsilon_0} (r^2 - a^2) - \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b}{a}\right) = V_b(0 < r \leq a) // \end{aligned}$$

P_2

Ⓟ a) Sí es un condensador porque es un arreglo de conductores.

Ⓟ b) Como solo se pide la capacitancia, se requiere una d.d.p. entre las placas.

Para calcular dicha d.d.p., se requiere el campo!

Por simetría cilíndrica, se usa ley de Gauss entre las placas, asumiendo que existe una carga en una de las armaduras:

$$a < r < b$$

$$\Rightarrow \oiint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

$$\Rightarrow \int_0^L \int_0^{2\pi} E(r) \hat{r} \cdot r d\theta dz \hat{r} = \frac{Q}{\epsilon_0}$$

$$\Rightarrow 2\pi r L E(r) = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E(r) = \frac{Q}{2\pi\epsilon_0 L} \frac{1}{r}$$

$$\Rightarrow \vec{E}(r) = \frac{Q}{2\pi\epsilon_0 L} \frac{1}{r} \hat{r}$$

→ Se calcula la d.d.p.:

$$V(a) - V(b) = - \int_a^b \vec{E} \cdot d\vec{r}$$

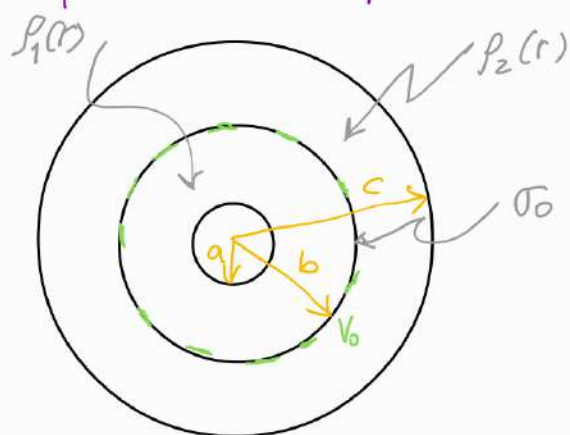
$$\Rightarrow \Delta V = \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)$$

$$\Rightarrow \frac{2\pi\epsilon_0}{\ln\left(\frac{b}{a}\right)} = \frac{Q}{\Delta V L}$$

justamente es capacitancia
unidad de largo //

P₃

Esquema del problema: (Resolución: muy similar a Φ_i)



Se conoce el campo eléctrico, que es definido por partes, para esta distribución de simetría cilíndrica.

Se tiene como dato el valor del potencial en $p=b$, $V(b)=V_0$.

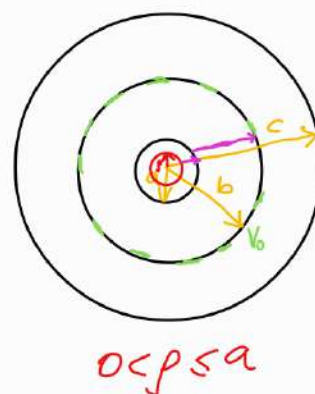
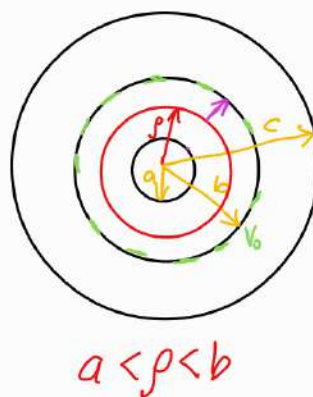
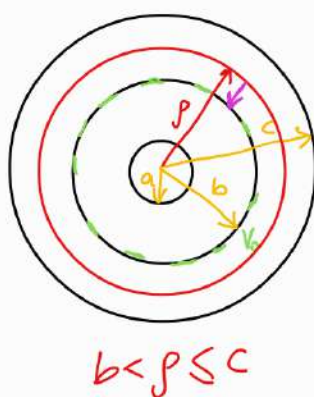
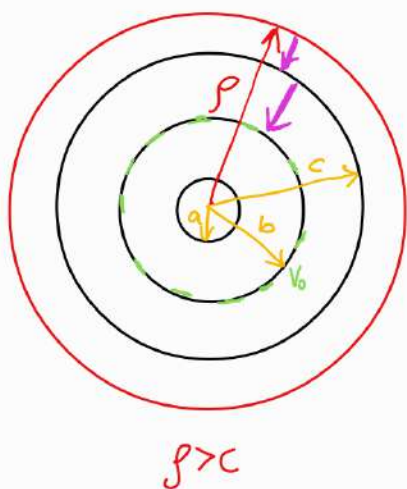
Para calcular el potencial en todo el espacio, siempre se requiere una referencia, tal que nos sea útil.

Como se conoce el valor del potencial ahí, se tomará esa ref.

Ya que el dato es el campo eléctrico, para llegar al potencial se utiliza $V(A) - V(B) = - \int_B^A \vec{E} \cdot d\vec{L}$.

Debido a que el campo eléctrico está definido por partes, el potencial también (en los mismos espacios), pero ¡DEBE ser continuo!

La idea siempre es "acercarse" a la referencia. Al hacer eso, se podría pasar por zonas con distinto campo eléctrico; por eso hay que considerarlo.



El campo eléctrico tiene la forma:

$$\vec{E} = \begin{cases} \frac{a^2}{\epsilon_0} \hat{p}, & 0 < p \leq a \\ \frac{a^2}{\epsilon_0} \frac{1}{p} \hat{p}, & a < p < b \\ \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} \right) \frac{1}{p} + \underbrace{\frac{bc}{\epsilon_0} \frac{1}{p} \ln\left(\frac{p}{b}\right)}_{= \frac{bc}{\epsilon_0} \frac{1}{p} (\ln(p) - \ln(b))} \hat{p}, & b < p \leq c \\ \underbrace{\left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \frac{1}{p} + \frac{bc}{\epsilon_0} \frac{1}{p} \ln(p)}_{= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \frac{1}{p} \hat{p}}, & p > c \end{cases}$$

$\mu = \text{cte}$

para facilitar
un poco la
matraca

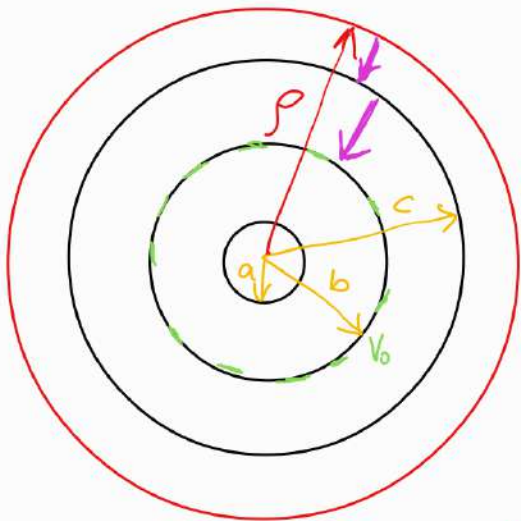


Si tomamos $r_{\text{ref}} = \infty$ y $V(r_{\text{ref}}) = 0$:

$$\begin{aligned} V(p > c) - V(\infty) &= - \int_{\infty}^p \vec{E} \cdot d\vec{\ell} = - \int_{\infty}^p \vec{E}(p > c) \cdot d\vec{\ell} \\ &= - \mu \int_{\infty}^p \frac{1}{p} dp = - \mu [\ln(p)]_{\infty}^p = - \mu [\ln(p) - \underbrace{\ln(p \rightarrow \infty)}_{\rightarrow \infty}] \end{aligned}$$

Moraleja: como cilindro no está acotado en el espacio, siempre tomamos la referencia en algún radio conocido.

$$\underline{p > c}$$



$$\bullet V(p > c) - V(b) = - \int_b^p \vec{E} \cdot d\vec{l}$$

$$= - \left(\underbrace{\int_c^p \vec{E}(p > c) \cdot d\vec{l}}_{\text{I}} + \underbrace{\int_b^c \vec{E}(b < p \leq c) \cdot d\vec{l}}_{\text{II}} \right)$$

(ver desarrollo
más abajo)

$$= - \left[\underbrace{\left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \ln\left(\frac{p}{c}\right)}_{\text{I}} + \underbrace{\left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{c}{b}\right)}_{\text{II}} \right]$$

$$\Rightarrow V(p > c) = - \left[\left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \ln\left(\frac{p}{c}\right) + \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{c}{b}\right) \right] + V_0$$

$$\textcircled{\text{I}}_{p>c}: \int_c^p \vec{E}(p>c) \cdot d\vec{\ell} \quad \left\| \begin{array}{l} \cdot \vec{E}(p>c) = \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \frac{1}{p} \hat{p} \\ \cdot d\vec{\ell} = dp \hat{p}, \quad p>c \end{array} \right.$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \int_c^p \frac{1}{p} dp$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \left[\ln(|p|) \right]_c^p \quad ; \quad |p|>0 \Rightarrow |p|=p$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \left[\ln(p) - \ln(c) \right]$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \ln\left(\frac{p}{c}\right) //$$

II: $\oint_{p>c} \vec{E}(b < p \leq c) \cdot d\vec{\ell} \quad \left| \begin{array}{l} \cdot \vec{E}(b < p \leq c) = \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \frac{1}{p} + \frac{bc}{\epsilon_0} \frac{1}{p} \ln(p) \\ \cdot d\vec{\ell} = dp \hat{p} \end{array} \right.$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \int_b^c \frac{1}{p} dp + \frac{bc}{\epsilon_0} \int_b^c \frac{1}{p} \ln(p) dp \quad \left. \begin{array}{l} \cdot u = \ln(p) \quad \left| \begin{array}{l} p=c \Rightarrow u=\ln(c) \\ p=b \Rightarrow u=\ln(b) \end{array} \right. \\ \Rightarrow du = \frac{1}{p} \end{array} \right\}$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \left[\ln(|p|) \right]_b^c + \frac{bc}{\epsilon_0} \int_{\ln(b)}^{\ln(c)} u du \quad ; \quad p > 0 \Rightarrow |p| = p$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) [\ln(c) - \ln(b)] + \frac{bc}{\epsilon_0} \cdot \frac{1}{2} [u^2]_{\ln(b)}^{\ln(c)}$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) [\ln(c) - \ln(b)] + \frac{bc}{2\epsilon_0} \underbrace{[\ln(c)^2 - \ln(b)^2]}_{[\ln(c) + \ln(b)][\ln(c) - \ln(b)]}$$

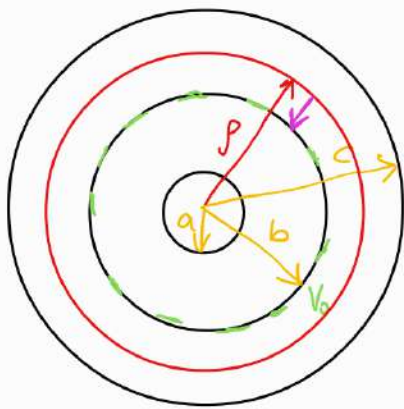
$$= [\ln(c) - \ln(b)] \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) + \frac{bc}{2\epsilon_0} [\ln(c) + \ln(b)] \right)$$

$$= \frac{bc}{2\epsilon_0} \ln(c) + \frac{bc}{2\epsilon_0} \ln(b)$$

$$= \ln\left(\frac{c}{b}\right) \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(b) + \frac{bc}{2\epsilon_0} \ln(c) \right)$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{c}{b}\right)$$

$$\underline{b < p \leq c}$$



$$\bullet V(b < p \leq c) - V(b) = - \int_b^p \vec{E} \cdot d\vec{r}$$

$$= - \left(\int_b^p \vec{E}(b < p \leq c) \cdot d\vec{r} \right)$$

$\textcircled{\text{I}}$
 $b < p \leq c$

(ver desarrollo más abajo)

$$= \underbrace{\left(\frac{a^2}{\epsilon_0} + \frac{q_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bp) \right)}_{\textcircled{\text{I}} \atop b < p \leq c} \ln\left(\frac{p}{b}\right)$$

$$\Rightarrow V(b < p \leq c) = - \ln\left(\frac{p}{b}\right) \left(\frac{a^2}{\epsilon_0} + \frac{q_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bp) \right) + V_0 //$$

$\textcircled{II}$
 $p > c$
 ss

$\textcircled{I}$:
 $b < p < c$

$$\int_b^p \vec{E}(b < p < c) \cdot d\vec{r} \quad \left\| \begin{array}{l} \cdot \vec{E}(b < p < c) = \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \frac{1}{p} + \frac{bc}{\epsilon_0} \frac{1}{p} \ln(p) \\ \cdot d\vec{r} = dp \hat{r} \end{array} \right.$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \int_b^p \frac{1}{p} dp + \frac{bc}{\epsilon_0} \int_b^p \frac{1}{p} \ln(p) dp \quad \left\{ \begin{array}{l} \cdot u = \ln(p) \quad \left\| \begin{array}{l} p=p \Rightarrow u=\ln(p) \\ p=b \Rightarrow u=\ln(b) \end{array} \right. \\ \Rightarrow du = \frac{1}{p} \end{array} \right.$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) \left[\ln(p) \right]_b^p + \frac{bc}{\epsilon_0} \int_{\ln(b)}^{\ln(p)} u du \quad ; \quad p > 0 \Rightarrow |p| = p$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) [\ln(p) - \ln(b)] + \frac{bc}{\epsilon_0} \cdot \frac{1}{2} [u^2]_{\ln(b)}^{\ln(p)}$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) \right) [\ln(p) - \ln(b)] + \frac{bc}{2\epsilon_0} \underbrace{[\ln(p)^2 - \ln(b)^2]}_{[\ln(p) + \ln(b)][\ln(p) - \ln(b)]}$$

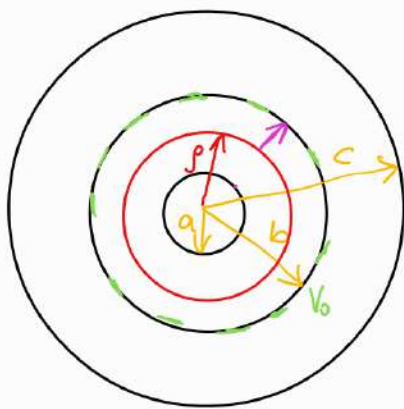
$$= [\ln(p) - \ln(b)] \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} - \frac{bc}{\epsilon_0} \ln(b) + \frac{bc}{2\epsilon_0} [\ln(p) + \ln(b)] \right)$$

$$= \underbrace{\frac{bc}{2\epsilon_0} \ln(p)} + \underbrace{\frac{bc}{2\epsilon_0} \ln(b)}$$

$$= \ln\left(\frac{p}{b}\right) \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(b) + \frac{bc}{2\epsilon_0} \ln(p) \right)$$

$$= \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bp) \right) \ln\left(\frac{p}{b}\right)$$

$$\underline{a < p \leq b}$$



$$\begin{aligned} \bullet V(a < p < b) - V(b) &= - \int_b^p \vec{E} \cdot d\vec{r} \\ &= - \underbrace{\left(\int_b^p \vec{E}(a < p < b) \cdot d\vec{r} \right)}_{\substack{\textcircled{I} \\ a < p \leq b}} \quad (\text{ver desarrollo} \\ &\quad \text{más abajo}) \end{aligned}$$

$$= - \underbrace{\frac{a^2}{\epsilon_0} \ln\left(\frac{p}{b}\right)}_{\substack{\textcircled{I} \\ a < p \leq c}}$$

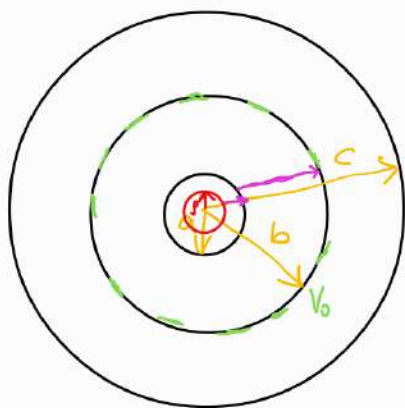
$$\Rightarrow V(b < p \leq c) = -\frac{a^2}{\epsilon_0} \ln\left(\frac{p}{b}\right) + V_0 //$$

$$\textcircled{I} : \int_b^p \vec{E}(a < \rho < b) \cdot d\vec{\ell} \quad \left\| \begin{array}{l} \cdot \vec{E}(a < \rho < b) = \frac{a^2}{\epsilon_0} \frac{1}{\rho} \hat{\rho} \\ \cdot d\vec{\ell} = d\rho \hat{\rho}, \quad a < \rho \leq b \end{array} \right.$$

$$= \frac{a^2}{\epsilon_0} \int_b^p \frac{1}{\rho} d\rho = \frac{a^2}{\epsilon_0} \left[\ln(|\rho|) \right]_b^p = \frac{a^2}{\epsilon_0} [\ln(p) - \ln(b)] = \frac{a^2}{\epsilon_0} \ln\left(\frac{p}{b}\right)$$

$\rho > 0 \Rightarrow |\rho| = \rho$

$$0 < \rho \leq a]$$



$$\bullet V(0 < \rho \leq a) - V(b) = - \int_b^{\rho} \vec{E} \cdot d\vec{\ell}$$

$$= - \left(\underbrace{\int_a^{\rho} \vec{E}(0 < \rho \leq a) \cdot d\vec{\ell}}_{\textcircled{I} \atop 0 < \rho \leq a} + \underbrace{\int_b^a \vec{E}(a < \rho < b) \cdot d\vec{\ell}}_{\textcircled{II} \atop 0 < \rho \leq a} \right)$$

(ver desarrollo
más abajo)

$$= - \left(\underbrace{\left(\frac{a}{\epsilon_0} (\rho - a) \right)}_{\textcircled{I} \atop 0 < \rho \leq a} + \underbrace{\left(\frac{a^2}{\epsilon_0} \ln\left(\frac{a}{b}\right) \right)}_{\textcircled{II} \atop 0 < \rho \leq a} \right)$$

$$\Rightarrow V(0 < \rho \leq a) = - \left(\frac{a}{\epsilon_0} (\rho - a) + \frac{a^2}{\epsilon_0} \ln\left(\frac{a}{b}\right) \right) + V_0$$

$$\begin{aligned}
 \textcircled{\text{I}} : \int_a^{\rho} \vec{E}(0 < \rho \leq a) \cdot d\vec{\ell} \quad & \parallel \cdot \vec{E}(0 < \rho \leq a) = \frac{a}{\epsilon_0} \hat{\rho} \\
 0 < \rho \leq a \quad & \cdot d\vec{\ell} = d\rho \hat{\rho}, \quad 0 < \rho \leq a \\
 \\
 & = \frac{a}{\epsilon_0} \int_a^{\rho} d\rho = \frac{a}{\epsilon_0} [\rho]_a^{\rho} = \frac{a}{\epsilon_0} [\rho - a] //
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{\text{I}} \quad a < \rho < b \\
 \text{SS} \\
 \textcircled{\text{II}} : \int_b^a \vec{E}(a < \rho < b) \cdot d\vec{\ell} \quad & \parallel \cdot \vec{E}(a < \rho < b) = \frac{a^2}{\epsilon_0} \frac{1}{\rho} \hat{\rho} \\
 0 < \rho \leq a \quad & \cdot d\vec{\ell} = d\rho \hat{\rho}, \quad a < \rho < b \\
 \\
 & = \frac{a^2}{\epsilon_0} \int_b^a \frac{1}{\rho} d\rho = \frac{a^2}{\epsilon_0} \left[\ln(|\rho|) \right]_b^a = \frac{a^2}{\epsilon_0} [\ln(a) - \ln(b)] = \frac{a^2}{\epsilon_0} \ln\left(\frac{a}{b}\right) // \\
 & \quad \rho > 0 \Rightarrow |\rho| = \rho
 \end{aligned}$$

Entonces, queda el potencial definido en todo el espacio, con respecto a la referencia en $\rho=b$:

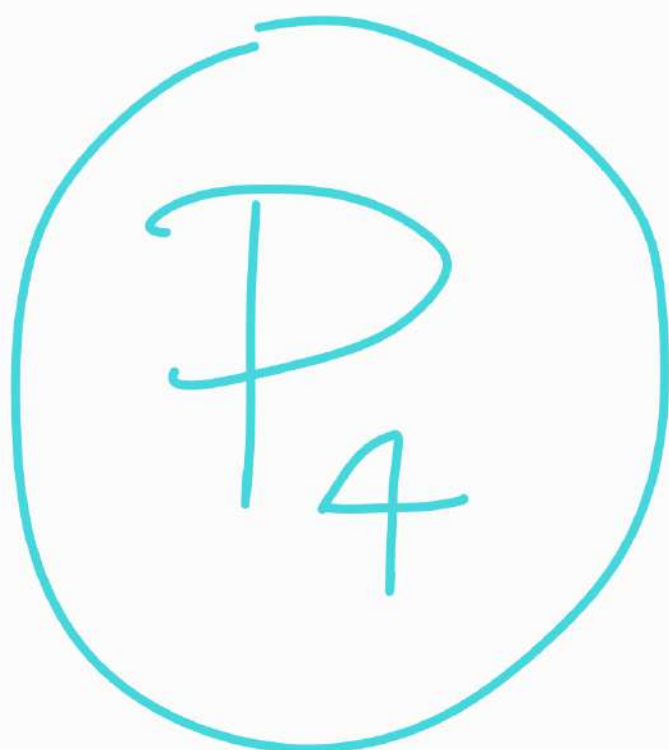
$$\left\{ \begin{aligned} \bullet V(0 < \rho \leq a) &= - \left(\frac{a}{\epsilon_0} (\rho - a) + \frac{a^2}{\epsilon_0} \ln\left(\frac{a}{b}\right) \right) + V_0 \\ \bullet V(a < \rho < b) &= - \frac{a^2}{\epsilon_0} \ln\left(\frac{\rho}{b}\right) + V_0 \\ \bullet V(b < \rho \leq c) &= - \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{\rho}{b}\right) + V_0 \\ \bullet V(\rho > c) &= - \left[\left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \ln\left(\frac{\rho}{c}\right) + \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{c}{b}\right) \right] + V_0 \end{aligned} \right.$$

y además, está bien definido pues es continuo:

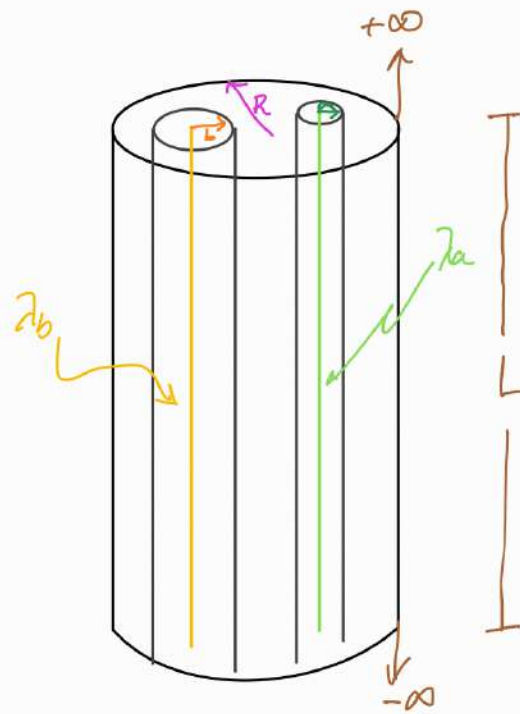
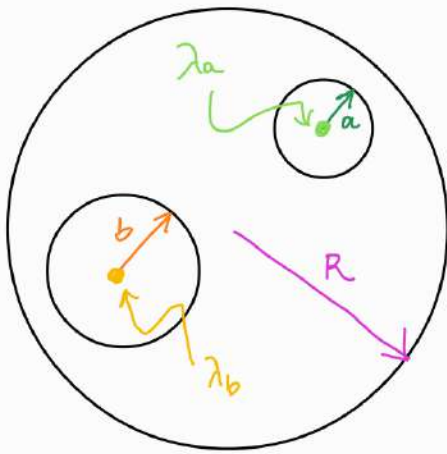
$$\begin{aligned} \bullet V(0 < \rho \leq a) \Big|_{\rho=a} &= - \left(\frac{a}{\epsilon_0} (a-a) + \frac{a^2}{\epsilon_0} \ln\left(\frac{a}{b}\right) \right) + V_0 \\ &= - \frac{a^2}{\epsilon_0} \ln\left(\frac{a}{b}\right) + V_0 = V(a < \rho < b) \Big|_{\rho=b} // \end{aligned}$$

$$\begin{aligned} \bullet V(a < \rho < b) \Big|_{\rho=b} &= - \frac{a^2}{\epsilon_0} \ln\left(\frac{b}{b}\right) + V_0 \\ &= - \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{b}{b}\right) + V_0 = V(b < \rho \leq c) \Big|_{\rho=b} // \end{aligned}$$

$$\begin{aligned} \bullet V(b < \rho \leq c) \Big|_{\rho=c} &= - \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{c}{b}\right) + V_0 \\ &= - \left[\left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{\epsilon_0} \ln\left(\frac{c}{b}\right) \right) \ln\left(\frac{c}{c}\right) + \left(\frac{a^2}{\epsilon_0} + \frac{\sigma_0 b}{\epsilon_0} + \frac{bc}{2\epsilon_0} \ln(bc) \right) \ln\left(\frac{c}{b}\right) \right] + V_0 \\ &= V(\rho > c) \Big|_{\rho=c} // \end{aligned}$$



Esquema del problema :



Ya que el conductor está descargado, tiene carga neta cero; pero esto no significa que no pueda tener carga en el exterior.

(Si estuviese cargado, la única forma de que tuviese carga es que esta se distribuya en la superficie, ya que dentro no puede encerrar carga pues el campo eléctrico es nulo dentro).

Sin embargo, las paredes internas de las cavidades de radio a y b también son conductores (pues son parte del cilindro macizo), luego si existe carga en esas cavidades, las superficies deben estar cargadas de manera tal que se anulen las cargas (y así, también se cumple que el campo eléctrico es nulo).

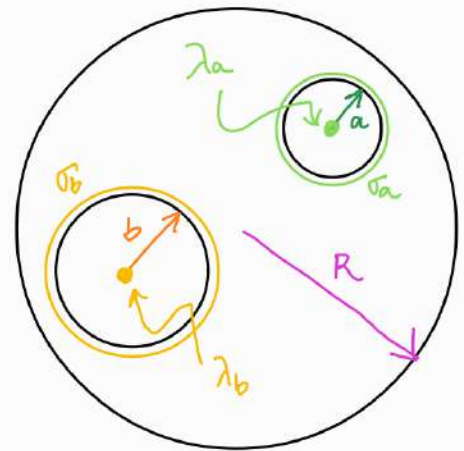
P_3 a) Ya que $(\text{densidad de carga lineal}) = \frac{(\text{carga})}{(\text{longo})}$, entonces en cada caso:

$$\begin{aligned} \bullet \lambda_b &= \frac{Q_b}{L} & \bullet \lambda_a &= \frac{Q_a}{L} \\ \Rightarrow Q_b &= \lambda_b L & \Rightarrow Q_a &= \lambda_a L \end{aligned} \left. \vphantom{\begin{aligned} \bullet \lambda_b &= \frac{Q_b}{L} \\ \bullet \lambda_a &= \frac{Q_a}{L} \end{aligned}} \right\} \begin{array}{l} \text{las cargas que se distribuyen en} \\ \text{la línea no tienen por qué ser} \\ \text{iguales (por eso se usan etiquetas)} \\ \text{(pero sí el largo, por la geometría)} \end{array}$$

Por otro lado, $(\text{densidad de carga superficial}) = \frac{(\text{carga})}{(\text{área})}$, entonces en cada manto entorno a las distribuciones lineales, se debe distribuir la carga opuesta a su respectivas distribuciones en el centro:

$$\begin{aligned} \bullet \sigma_b &= \frac{-Q_b}{2\pi b L} & \bullet \sigma_a &= \frac{-Q_a}{2\pi a L} \\ &= \frac{-\lambda_b L}{2\pi b L} & &= \frac{-\lambda_a L}{2\pi a L} \\ \Rightarrow \sigma_b &= \frac{-\lambda_b}{2\pi b} & \Rightarrow \sigma_a &= \frac{-\lambda_a}{2\pi a} \end{aligned}$$

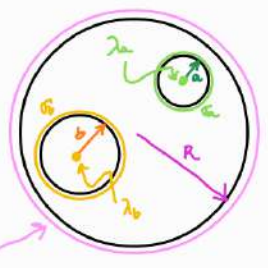
y entonces se tiene esto:



Pero debido a σ_b y σ_a , se está generando una carga neta negativa en el cilindro macizo. Pero como está descargado, esta carga no puede ir a esa superficie; luego debe haber otra carga que lo concibe.

$$\begin{aligned} \bullet \sigma_R &= \frac{-((-Q_a) + (-Q_b))}{2\pi R L} = \frac{Q_a + Q_b}{2\pi R L} \\ &= \frac{\lambda_a L + \lambda_b L}{2\pi R L} = \frac{(\lambda_a + \lambda_b)L}{2\pi R L} \end{aligned}$$

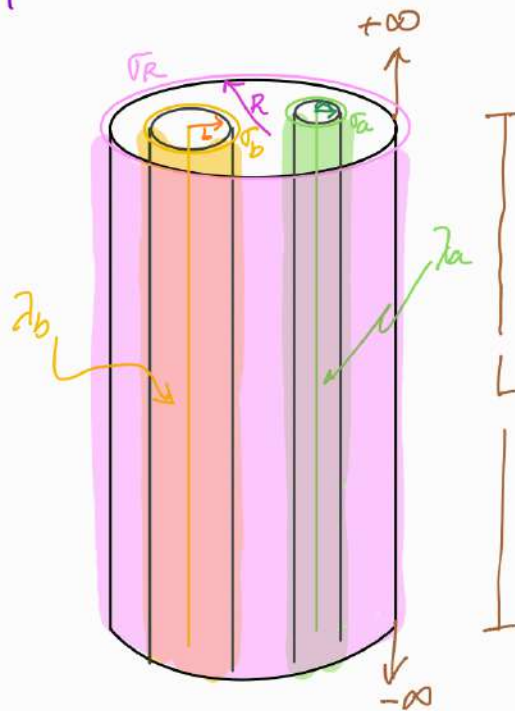
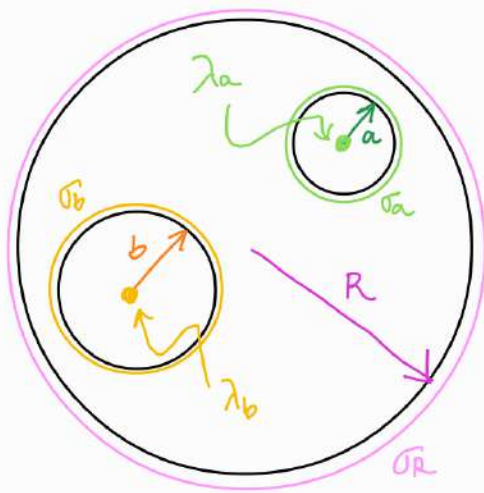
$$\Rightarrow \sigma_R = \frac{\lambda_a + \lambda_b}{2\pi R}$$



La densidad volumétrica de carga es nula! (no hay carga en total; es conductor).

y queda: σ_R

(P₃) b) De lo anterior, se obtuvo que:



Hay simetría cilíndrica $\leadsto \vec{E} = \vec{E}(\rho, \theta, z)$

* el sistema se ve igual al rotarlo $\Rightarrow \vec{E} = \vec{E}(\rho, z)$

* el sistema luce igual en toda altura $\Rightarrow \vec{E} = \vec{E}(\rho)$

* las líneas de campo van en $\hat{\rho} \Rightarrow \vec{E} = E(\rho)\hat{\rho}$

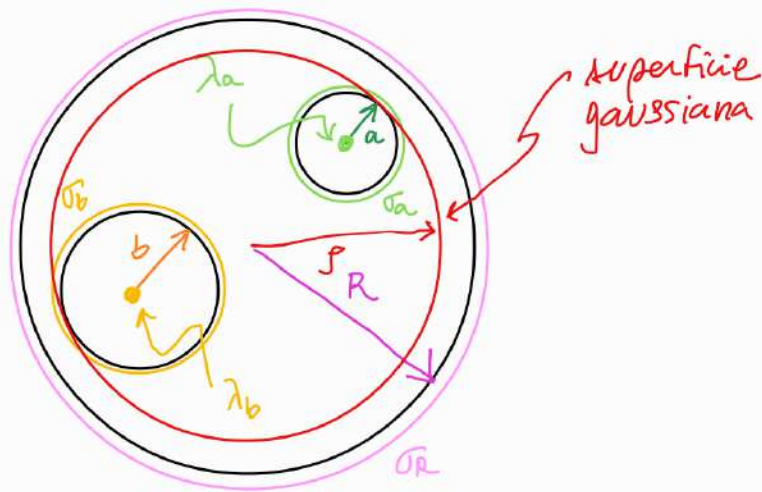
Entonces se puede usar ley de Gauss: $\oiint \vec{E} \cdot d\vec{S} = \frac{Q_{encerrada}}{\epsilon_0}$

El cilindro muestra diferencia 2 regiones:

(i) $0 < \rho < R$ (dentro)

(ii) $R < \rho$ (fuera)

(i) $0 < \rho < R$



L.I.1

$$\oint \vec{E} \cdot d\vec{S} \left| \begin{array}{l} \vec{E} = E(\rho) \hat{\rho} \\ d\vec{S} = \rho d\theta dz \hat{\rho} + \rho d\theta d\hat{z} + \rho d\rho d\hat{\theta} \end{array} \right| = E(\rho) \rho d\theta dz$$

$(0 < \rho < R)$
 $(0 < \theta < 2\pi)$
 $(0 < z < L)$

$$\Rightarrow \oint_0^{2\pi} \int_0^L E(\rho) \rho d\theta dz = E(\rho) \rho \int_0^L dz \int_0^{2\pi} d\theta = E(\rho) \rho [z]_0^L [\theta]_0^{2\pi}$$

$$= E(\rho) \rho \cdot [L-0] \cdot [2\pi-0] = E(\rho) \cdot 2\pi \rho L$$



superficie de un trazo de cilindro de largo L y radio ρ , entre 0 y R .

L.D.1

$$Q_{\text{encerrada}} = Q_a + Q_b + (-Q_a) + (-Q_b) = 0 \quad (\text{es un conductor!})$$

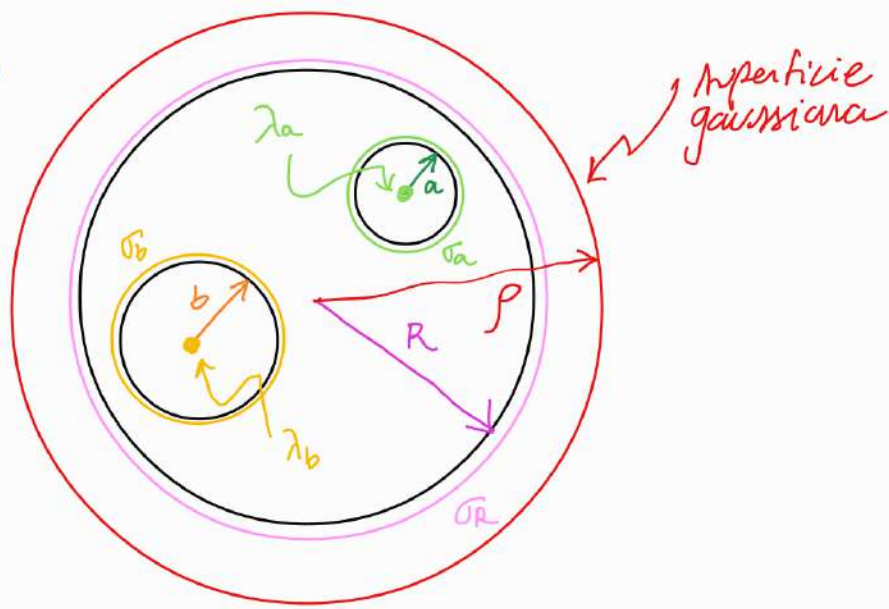
$$\Rightarrow \frac{Q_{\text{encerrada}}}{\epsilon_0} = 0$$

— 0 —

$$\Rightarrow E(\rho) \cdot 2\pi \rho L = 0$$

$$\Rightarrow \boxed{\vec{E}(\rho) = 0 \hat{\rho}}, \quad 0 < \rho < R \quad (\text{lo cual tiene mucho sentido pues es un conductor})$$

(ii) $R < \rho$



L.I.1

$$\oint \vec{E} \cdot d\vec{S} \quad \left| \begin{array}{l} \vec{E} = E(\rho) \hat{\rho} \\ d\vec{S} = \rho d\theta dz \hat{\rho} + \rho d\theta d\hat{z} + \rho d\rho d\theta \hat{z} \end{array} \right| = \vec{E} \cdot d\vec{S}$$

$(R \leq \rho)$
 $(0 < \theta < 2\pi)$
 $(0 < z < L)$

$$\Rightarrow \int_0^L \int_0^{2\pi} E(\rho) \rho d\theta dz = E(\rho) \rho \int_0^L dz \int_0^{2\pi} d\theta = E(\rho) \rho [z]_0^L [\theta]_0^{2\pi}$$

$$= E(\rho) \rho \cdot [L-0] \cdot [2\pi-0] = E(\rho) \cdot 2\pi \rho L$$



superficie de un tubo de cilindro de largo L y radio ρ , entre 0 y R .

L.D.1

$$\bullet Q_{encerrada} = \sigma_R \cdot 2\pi R L = \lambda_a + \lambda_b$$

$$\Rightarrow \frac{Q_{encerrada}}{\epsilon_0} = \frac{\lambda_a + \lambda_b}{\epsilon_0}$$

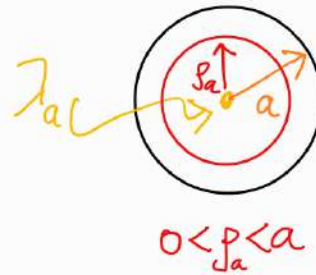
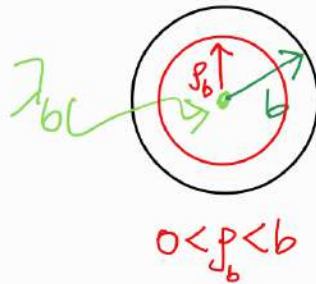
—○—

$$\Rightarrow E(\rho) \cdot 2\pi \rho L = \frac{\lambda_a + \lambda_b}{\epsilon_0}$$

$$\Rightarrow \vec{E}(\rho) = \frac{\lambda_a + \lambda_b}{2\pi \epsilon_0 L} \frac{1}{\rho} \hat{\rho} \quad , \quad \rho > R$$

P_3 c)

Si se pide en cada cavidad, entonces es dentro de un radio que no supere su borde (medido desde el centro respectivo):



Se puede usar Ley de Gauss por la justificación $\star$.
Para λ y radio r genéricos:

L.I.

$$\bullet \oint \vec{E} \cdot d\vec{S} \left\| \begin{array}{l} \vec{E} = E(r) \hat{r} \\ d\vec{S} = r d\theta dz \hat{r} + r dz d\theta \hat{\theta} + r dr d\theta \hat{z} \end{array} \right\| = E(r) \int_0^L \int_0^{2\pi} r d\theta dz$$

$(0 < r < a)$
 $(0 < \theta < 2\pi)$
 $(0 < z < L)$

$$\Rightarrow \int_0^L \int_0^{2\pi} E(r) r d\theta dz = E(r) \cdot 2\pi r L$$

L.D.

$$\bullet Q_{\text{encerrada}} = \lambda \cdot L$$

$$\Rightarrow \frac{Q_{\text{encerrada}}}{\epsilon_0} = \frac{\lambda \cdot L}{\epsilon_0}$$

— 0 —

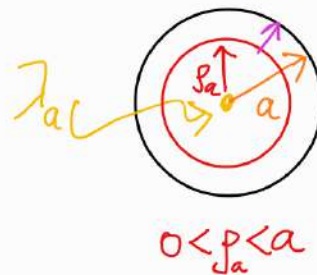
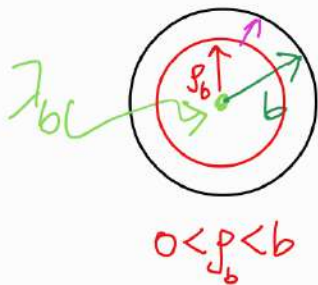
$$\Rightarrow E(r) \cdot 2\pi r L = \frac{\lambda L}{\epsilon_0} \Rightarrow \vec{E}(r) = \frac{\lambda}{2\pi\epsilon_0} \frac{1}{r} \hat{r}, \quad 0 < r < a$$

En particular, $\boxed{\vec{E}_{a,b}(r) = \frac{\lambda_{a,b}}{2\pi\epsilon_0} \frac{1}{r} \hat{r}}, \quad 0 < r_{a,b} < a,b$

(P3) d) Para calcular d.d.p., con $\vec{E}$ conocido, se usa:

$$\Delta V = V(A) - V(B) = - \int_B^A \vec{E} \cdot d\vec{\ell}$$

y se puede tomar como referencia el borde del conductor, pues es equipotencial (llámese $V_a(a) = V_b(b) = V_0$)



Para el caso de b, en detalle:

$$\begin{aligned} V_b(r_b < b) - V_b(b) &= - \int_b^{r_b} \vec{E} \cdot d\vec{\ell} \\ &= - \int_b^{r_b} \vec{E}_b(0 < r_b < b) \cdot d\vec{\ell} \quad \left| \begin{array}{l} \vec{E}_b(0 < r_b < b) = \frac{\lambda_b}{2\pi\epsilon_0} \frac{1}{r_b} \hat{r}_b \\ d\vec{\ell} = dr_b \hat{r}_b \end{array} \right. \cdot \vec{E}_b(0 < r_b < b) \cdot d\vec{\ell} \\ &= - \frac{\lambda_b}{2\pi\epsilon_0} \frac{1}{r_b} \int_b^{r_b} dr_b \end{aligned}$$

$$= - \frac{\lambda_b}{2\pi\epsilon_0} \int_b^{r_b} \frac{1}{r_b} dr_b = - \frac{\lambda_b}{2\pi\epsilon_0} [\ln(r_b)]_b^{r_b} = - \frac{\lambda_b}{2\pi\epsilon_0} [\ln(r_b) - \ln(b)]$$

$r_b > 0 \Rightarrow |r_b| = r_b$

$$\Rightarrow V_b(r_b < b) = - \frac{\lambda_b}{2\pi\epsilon_0} \ln\left(\frac{r_b}{b}\right) + V_0 //$$

Análogamente, $V_a(r_a < a) = - \frac{\lambda_a}{2\pi\epsilon_0} \ln\left(\frac{r_a}{a}\right) + V_0 //$

ánimo! ☺



⚠P.D. : En general, el desarrollo de los epurios no suele ser ~~taaaan~~ extenso; pero yo lo hice así aquí para no saltarme pasos! De a poco encontrarán su método.